

CENG381 Stochastic Processes

De Moivre's Martingale — Practice 1

Q1 (30 points). Consider a biased random walk with steps $X_i = +1$ w.p. p and $X_i = -1$ w.p. $q = 1-p$ ($p \neq 1/2$). The drift per step is $\mu = p - q = 2p - 1$. Define the compensated random walk $Y_n = S_n - n(p - q)$.

- Show that $\{Y_n; n \geq 0\}$ is a martingale. Write $Y_n = Y_{n-1} + (X_n - (p-q))$ and compute $E[X_n - (p-q)]$ to verify the martingale condition.
 - What is $E[S_n | S_0 = 0]$? What is $E[Y_n]$? Interpret Y_n as S_n with its expected growth removed.
 - For $p = 0.7$ (biased towards $+1$): (i) Compute the drift $\mu = p - q$. (ii) A gambler starts at $S_0 = 5$. Write down Y_n for this case and compute $E[S_{100} | S_0 = 5]$.
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Q2 (35 points). De Moivre's Martingale: for a biased random walk ($p \neq 1/2$) with $S_0 = k$, define $M_n = (q/p)^{S_n}$.

- Prove $\{M_n\}$ is a martingale. Use the factorization $M_n = M_{n-1} \cdot (q/p)^{X_n}$ and compute $E[(q/p)^{X_n}]$ (where $X_n = +1$ w.p. p and -1 w.p. q). Show the result equals 1.
 - A gambler starts with $\$k$ and plays until reaching $\$0$ (ruin) or $\$N$ (goal). Let J be the stopping time and $p_k = P(\text{reach } N | S_0 = k)$. Apply the Optional Stopping Theorem to M_n : $E[M_J] = M_0$. At time J , either $S_J = N$ (prob p_k) or $S_J = 0$ (prob $1-p_k$). Derive the formula for p_k .
 - Check: for the symmetric walk $p = 1/2$, the formula $p_k = [(q/p)^k - 1] / [(q/p)^N - 1]$ is indeterminate ($0/0$). Set $r = q/p$ and apply L'Hôpital's rule as $r \rightarrow 1$ to show that $p_k \rightarrow k/N$. Verify this matches the fair-martingale result from the previous topic.
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Q3 (35 points). Apply De Moivre's formula $p_k = [(q/p)^k - 1] / [(q/p)^N - 1]$ to the following Gambler's Ruin problems.

- Favorable game: $p = 2/3$, $q = 1/3$. Gambler starts at $\$2$ with goal $\$4$ ($N = 4$). Compute q/p , then find p_2 . Sanity check: is $p_2 > 1/2$? (It should be, since $p > 1/2$.)
- Unfavorable game: $p = 1/3$, $q = 2/3$. Gambler starts at $\$3$ with goal $\$5$ ($N = 5$). Compute q/p , then find p_3 . Express p_3 as a fraction and check $p_3 < 1/2$.
- For the unfavorable game in (b), also compute p_1 and p_4 . Verify that p_k is a decreasing function of k (more money does not help when the game is unfavorable). Hint: check whether

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$dp_k/dk < 0$ or simply compare p_1, p_3, p_4 numerically.